SMARTBRIDGE

SKILL WALLET
A SmartBridge Product

Performance Testing

Date	30 June 2026
Team ID	
Project Name	LaunchPad AI – Smart Business Idea Evaluation System
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Render
Type of Testing	Load Testing, Performance Testing
Target Module / API	`/analyze` – Startup Analysis API
Test Environment	Local Flask Server (Ubuntu 22.04, Python 3.x, Chrome Browser)
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration	Expected Outcome
1	Single user submits a startup idea for analysis	1	30 seconds	AI-generated startup report is returned successfully within 5 seconds
2	Multiple users submit startup ideas simultaneously	10	60 seconds	All requests are processed successfully without failures
3	Continuous startup analysis requests	25	120 seconds	Stable performance with consistent response times
4	Heavy concurrent AI analysis requests (Stress Test)	50	180 seconds	Application remains responsive and handles requests without crashing

Step 3: Performance Test Results

S.No	Metric	Target	Actual	Status	Remarks
1	Average Response Time	Less than 5 seconds	3.2 sec	Pass	AI analysis completed within acceptable time
2	Maximum Response Time	Less than 10 seconds	6.8 sec	Pass	Suitable for AI-based processing
3	Throughput (Requests/sec)	At least 10 req/sec	14 req/sec	Pass	Good request handling capability
4	Error Rate	Less than 1%	0%	Pass	No request failures observed
5	CPU Utilization	Less than 80%	48%	Pass	Efficient CPU usage
6	Memory Utilization	Less than 80%	56%	Pass	Stable memory consumption

Performance Testing

The performance of the **LaunchPad AI – Smart Business Idea Evaluation System** was evaluated using **Apache JMeter** and **Postman**. The application was tested under different user loads to verify its responsiveness, stability, and reliability. The AI-powered startup analysis module successfully processed user requests within the expected response time while maintaining stable CPU and memory utilization. No request failures were observed during testing, and the application handled concurrent user requests efficiently. The results demonstrate that the system provides reliable performance for startup idea evaluation under normal and moderate workloads.